CS351 - IT Workshop II (Cloud Computing)

Assignment 1

Due on 10th August, 2026

Solve the following problems in Python. Write the programs with proper comments.

Problem 1:
Write a function to break down a string into a list of characters.

Input: "abc"
Output: ['a','b','c']

Problem 2:
Write a function to reverse output of the problem 1 back into a string.

Input: ['a','b','c']
Output: "abc"

Problem 3:
Write a function to generate a list of n random numbers. Use the inbuilt `random` module.

Input: 5
Output: [5,2,3,1,5]

Problem 4:
Write a function to sort a given list of numbers in descending order.

Input: [1,2,3,4,5]
Output: [5,4,3,2,1]

Problem 5:
Write a function to get the frequency of each number in a list of numbers. Use a python `dict` to solve this.

Input: [1,1,3,2,3,2,3,2,2]
Output: {1: 2, 3: 3, 2: 4}

Problem 6:
Write a function to get all the unique elements from a given list. Your solution must use `set` to solve this.

Input: [1,1,3,2,3,2,3,2,2]
Output: {1,2,3}

Problem 7:
Write a function to get the first repeating element from a list. Your solution must use `set` to solve this.

Input: [1,2,3,4,5,1,2]
Output: 1

Problem 8:
Write a function that takes an integer n and output a `dict` containing keys from 0,2 ... to n and each key is mapped to a list containing the square and cube of the number.

Input: 3
Output:
{
0:[0,0],
1:[1,1],
2:[4,8],
3:[9,27]
}

Problem 9:
Given two lists of equal size, write a function to create tuples of each consecutive element having the same index. Use `zip` in some capacity to solve this.

Input: [1,2,3,4], ['a','b','c','d']
Output: [(1,'a'), (2,'b'), (3,'c'), (4,'d')]

Problem 10:
Write a function that uses list comprehension to generate the squares of 0 to n.

Input : 5
Output : [0, 1, 4, 9, 16, 25]

Problem 11:
Write a function that uses dictionary comprehension to generate a mapping from (0 to n) to their squares.

Input : 5
Output : {0:0, 1:1, 2:4, 3:9, 4:16, 5:25}

Problem 12:
Write a `class` such that :
1. The initializer takes an arbitrary list of atomic values as input and saves it in an instance variable.
2. Has a method called `apply` which has the following functionality:
- Accepts a function as a parameter. You can use a lambda function.
- Applies the function to the saved list and returns the output. The instance variable must not be modified.
- If it fails `raise` an `Exception` with a custom error message. You can use `try` and `except` here.

Problem 13:
Write a function that takes as input a list of words and upper-cases each word. Use `functools.map` in some capacity to solve this.

Input : ['aa','bb','cd','e']
Output : ['AA', 'BB', 'CD', 'E']

Problem 14:
Write a function to find the product of all the numbers in a list using `functools.reduce` in some capacity.

Input : [1,2,3,4,5]
Output : 120